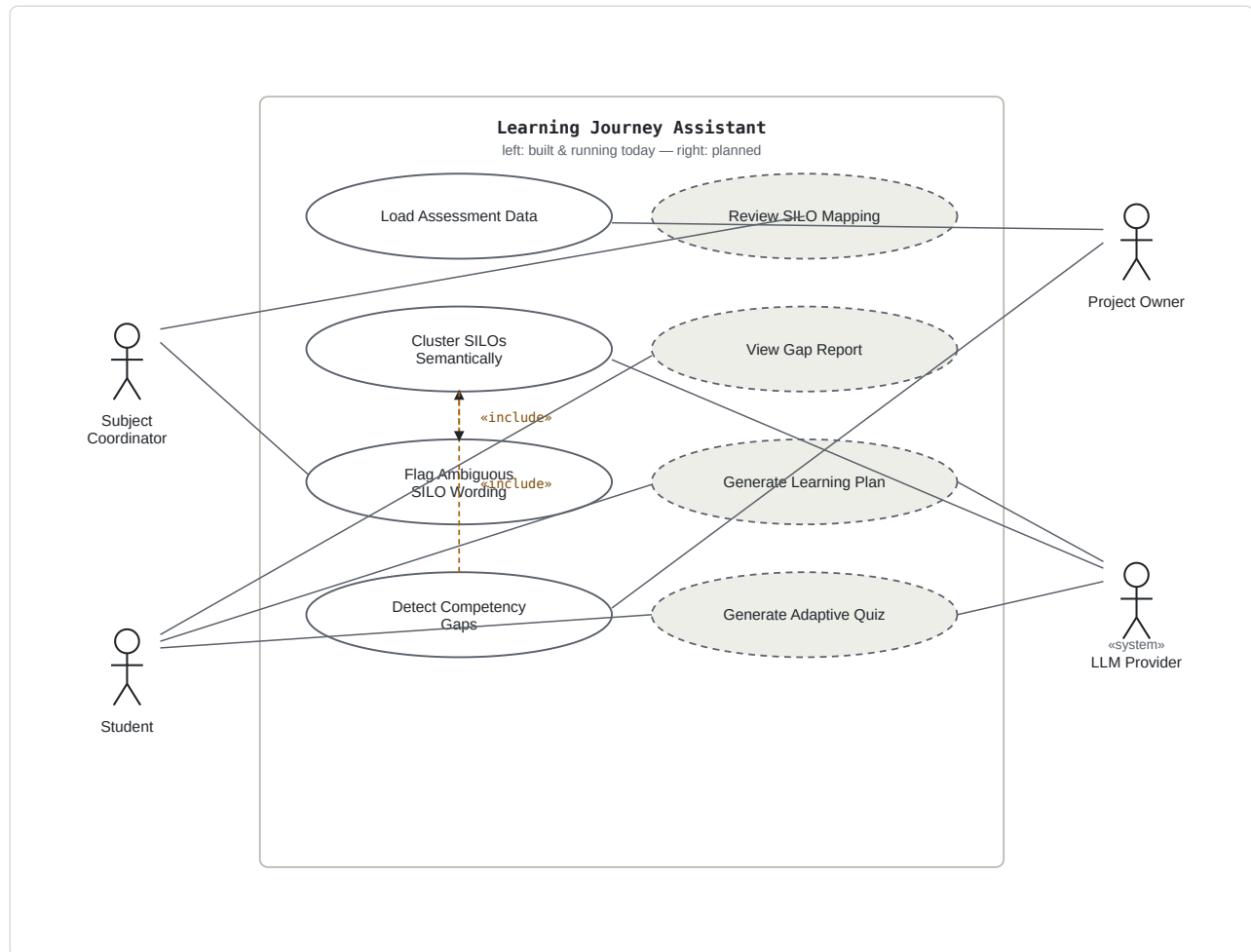


Use case and sequence diagrams

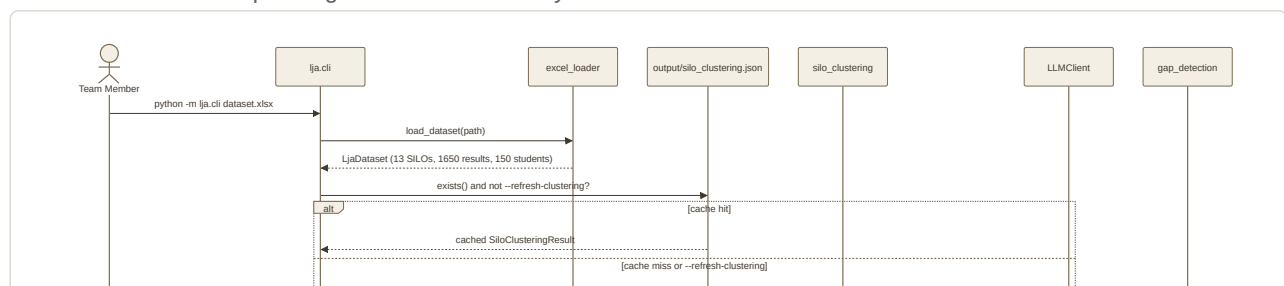
Companion to the [pipeline architecture diagram](#), which shows the data flowing through the system. This page shows the other two views UML asks for: **who** interacts with the system and what they can ask of it, and **what actually happens**, **message by message**, for the two flows that exist as running code today — the CLI pipeline, and the coverage-validation failure it was built to catch.

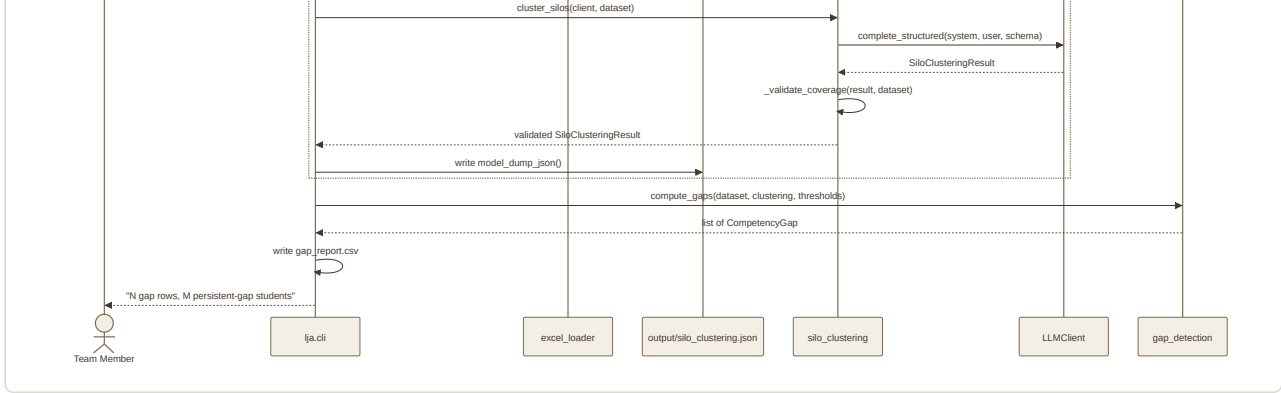


The left column is not aspirational — Load Assessment Data through Detect Competency Gaps ran end-to-end against the real 150-student dataset in this session, found 6 students with a genuine persistent gap, and is covered by 18 passing tests. The right column is exactly what's missing: nothing renders the gap report to the Student yet, no one has confirmed the LLM's SILO clustering, and no generation features exist on top of it.

SEQUENCE — RUNNING THE PIPELINE

What `python -m lja.cli dataset.xlsx` actually does, including the cache branch that avoids re-spending an LLM call on every run.

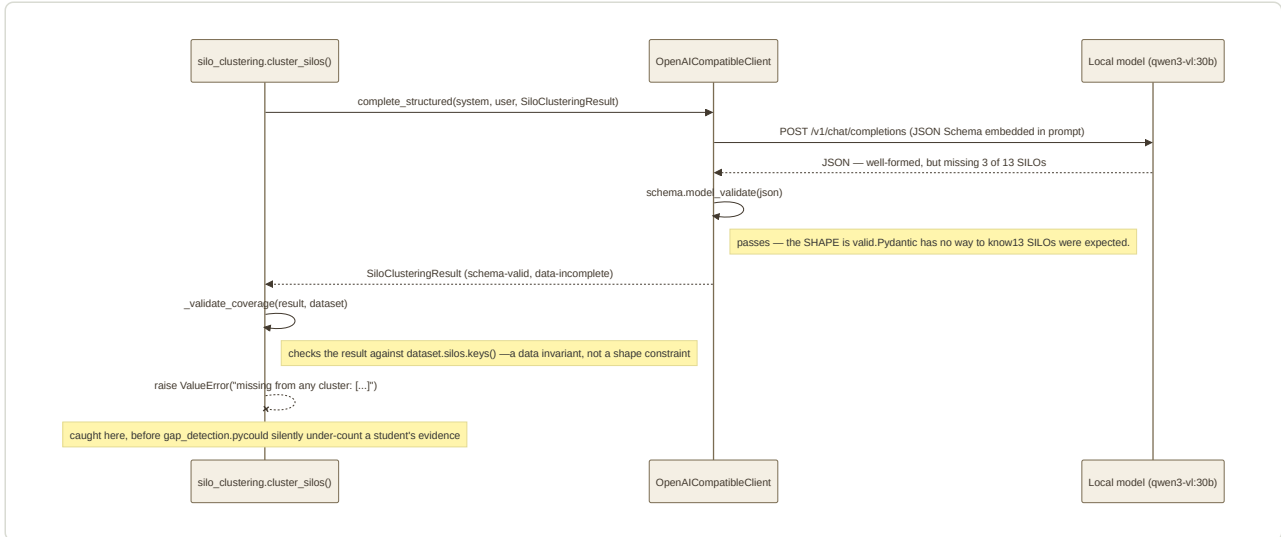




Run for real against qwen3-vl:30b : 1050 gap rows, 6 persistent-gap rows
 across 6 students, all in "Data Structures Knowledge and Application" — the same
 shape of cross-subject link Scott described on the call.

SEQUENCE – THE FAILURE THIS SESSION ACTUALLY CAUGHT

Why `_validate_coverage()` exists as a separate step, not folded into Pydantic's schema validation. A live run produced a response that was perfectly *shaped* — valid JSON matching the schema — while silently missing 3 of 13 SILOs. That's a data-completeness problem, which schema validation cannot see; only checking the response against the input dataset can.



Both sequence diagrams describe code that ran in this session against the real dataset, not a planned design. See `python/lja/model/silo_clustering.py` and `python/README.md`'s "real finding" section for the full account, including the prompt-ambiguity bug this validator also caught on a later run.